

Joaquin Garcia-Suarez

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Profile

Brief research statement

I am a mechanician with industry experience who uses theory, high-performance computing, and data-driven methods to tackle problems in interface mechanics, wave propagation, and their numerical modeling. My broad training in physics and applied mathematics enables original work focused on developing predictive tools to be deployed in practical settings. In the words of Prof. Hans Liepmann: “understanding of fundamentals with an appreciation for applications.”

Main interests

- ▶ **Contact problems**
Fluid-solid interaction (FSI), lubrication theory, friction modeling, wear prediction
- ▶ **Computational mechanics**
Data-driven methods, GPU adaptation
- ▶ **Acoustic, elastic and optical metamaterials**
Wave propagation in layered media, algebraic approaches

Education

- ▶ **Ph.D. Aeronautics (minor in Applied and Computational Mathematics)**, June 2020
California Institute of Technology (Pasadena, CA, USA)
Committee: Prof. D. Asimaki, Prof. M. Ortiz, Prof. G. Ravichandran, Prof. D. Meiron
Thesis: “Application of path-independent integrals to soil-structure interaction”
Defense date: September 27th 2019
- ▶ **M.S. Space Engineering**, June 2016
California Institute of Technology (Pasadena, CA, USA)
Adviser: Prof. M. Ortiz
- ▶ **“Ingeniero Industrial” diploma**, February 2013
Universidad de Sevilla (Sevilla, Spain)

Employment

Academia

- ▶ **École Polytechnique Fédérale de Lausanne (EPFL)**
 - SNSF Ambizione Fellow (Lausanne, Vaud, Switzerland), January 2024 – Present
 - Postdoctoral scientist at the Computational Solid Mechanics Laboratory (LSMS), advised by Prof. J.-F. Molinari (Lausanne, Vaud, Switzerland), April 2021 – September 2023

► **California Institute of Technology (Caltech)**

- Postdoctoral scholar at the Mechanical and Civil Engineering Department, advised by Prof. D. Asimaki (Pasadena, CA, USA), January 2020 – February 2021 (this extended stage was due to extraordinary sanitary situation)
- Graduate research assistant (Pasadena, CA, USA), July 2016 – December 2019

Industry _____

► **NextForce Engineering**

Structural engineer, tasked with design and FEM analysis of concentrated solar power structures (Sevilla, Spain), March 2013 – June 2015

Publications

Published peer-reviewed articles _____

*Corresponding author publications are marked with * by my name.*

- 24. **Garcia-Suarez, J.*** “A matter of shape: contact area optimization in soft lubrication”, *Tribology Letters* (2026). DOI: 10.1007/s11249-026-02108-1
- 23. Cortes, G., Sangiorgio, N., **Garcia-Suarez, J.*** “Phase-space iterative solvers”, *Computational Mechanics* (2025). DOI: 10.1007/s00466-025-02735-w
- 22. Gonzalez-Carbajal, J., Lemm, M., & **Garcia-Suarez, J.** “On the lowest-frequency bandgap of 1D phononic crystals”, *European Journal of Mechanics – A/Solids* (2025), DOI: 10.1016/j.euromechsol.2024.105466
- 21. Cortes, G., & **Garcia-Suarez, J.*** “Data-driven dynamic friction models based on recurrent neural networks”, *Applied Computing and Geosciences* (2025), DOI: 10.1016/j.acags.2025.100249
- 20. **Garcia-Suarez, J.**, McPhillips, D., Asimaki, D. “Seismic response of rock towers at the Trona Pinnacles (USA) to the 2019 Ridgecrest earthquake sequence: Theory, observations, and models”, *Bulletin of the Seismological Society of America* (2025), DOI: 10.1785/0120250073
- 19. Bilotto, J., Kolinski, J., Lecampion, B., Molinari, J.-F., Subhash, G., & **Garcia-Suarez, J.*** “Fluid-mediated impact of soft solids”, *Journal of Fluid Mechanics* (2024), DOI: 10.1017/jfm.2024.820
- 18. Voisin–Leprince, M., **Garcia-Suarez, J.**, Anciaux, G., & Molinari, J.-F. “Two-scale concurrent simulations for crack propagation using FEM/DEM bridging coupling”, *Computational Particle Mechanics* (2024), DOI: 10.1007/s40571-024-00788-x
- 17. **Garcia-Suarez, J.***, Brink, T., & Molinari, J.-F. “Roughness evolution induced by third-body wear”, *Tribology Letters* (2024), DOI: 10.1007/s11249-024-01833-9
- 16. **Garcia-Suarez, J.***, Brink, T., & Molinari, J.-F. “Breakdown of Reye’s theory in nanoscale wear”, *Journal of the Mechanics and Physics of Solids* (2023), DOI: 10.1016/j.jmps.2023.105236

- 15. **Garcia-Suarez, J.***, Cornet, A., Wattel, S., & Molinari, J.-F. “Data-driven 1D wave propagation for site response analysis”, *International Journal for Numerical and Analytical Methods in Geomechanics* (2023), DOI: 10.1002/nag.3596
- 14. Subhash, G., Bavdekar, S., Leonard III, R., Bagadion, M., **Garcia-Suarez, J.**, Cheenady, A., Sheplak, M., & Whittington, M. “Concept Article: A Novel Compact Millipede Bar Waveguide for Propagation of Longitudinal Stress Waves”, *Journal of Dynamic Behavior of Materials* (2023), DOI: 10.1007/s40870-023-00399-2
- 13. Voisin–Leprince, M., **Garcia-Suarez, J.**, Anciaux, G., & Molinari, J.-F. “FEM-DEM bridging coupling for the modeling of gouge”, *International Journal for Numerical Methods in Engineering* (2023), DOI: 10.1002/nme.7171
- 12. Wattel, S., Molinari, J.-F., Ortiz, M., & **Garcia-Suarez, J.*** “Mesh d-refinement: a data-based computational framework to account for complex material response”, *Mechanics of Materials* (2023), DOI: 10.1016/j.mechmat.2023.104630
- 11. **Garcia-Suarez, J.*** “Harmonic decomposition of the trace of 1D transfer matrices in layered media”, *Journal of the Mechanics and Physics of Solids* (2022), DOI: 10.1016/j.jmps.2022.104830
- 10. Wattel, S., **Garcia-Suarez, J.**, & Molinari, J.-F. “Understanding the mechanisms of adhesive wear for heterogeneous materials through atomistic simulations”, *Extreme Mechanics Letters* (2022), DOI: 10.1016/j.eml.2022.101913
- 9. **Garcia-Suarez, J.***, González Carbajal, J., & Asimaki, D. “Analytical 1D transfer functions for layered soils”, *Soil Dynamics and Earthquake Engineering* (2022), DOI: 10.1016/j.soildyn.2022.107532
- 8. Subhash, G., **Garcia-Suarez, J.**, Cheenady, A. A., Bavdekar, S., Whittington, W., & Molinari, J.-F. “Elastic stress wave propagation through a 180° bend in a square cross-section bar”, *International Journal of Engineering Science* (2022), DOI: 10.1016/j.ijengsci.2022.103748
- 7. **Garcia-Suarez, J.***, Seylabi, E., & Asimaki, D. “Application of ray methods to 1D site response of inhomogeneous soil deposits”, *Géotechnique* (2022), DOI: 10.1680/jgeot.21.00164
- 6. Donnellan, A., **Garcia-Suarez, J.**, McPhillips, D., Asimaki, D., Goulet, C., Meng, X., Devine, S. & Lyzenga, G. “Toppling of a Trona Pinnacles spire following the M5.5 Ridgecrest aftershock of June 2020”, *Seismological Research Letters* (2022), DOI: 10.1785/0220210275
- 5. **Garcia-Suarez, J.***, Asimaki, D., & Ortiz, M. “Applications of the J-integral to dynamical problems in geotechnical engineering”, *Journal of the Mechanics and Physics of Solids* (2021), DOI: 10.1016/j.jmps.2021.104353
- 4. **Garcia-Suarez, J.***, Seylabi, E., & Asimaki, D. “Seismic harmonic response of inhomogeneous soil: scaling analysis”, *Géotechnique* (2021), DOI: 10.1680/jgeot.19.P.042
- 3. **Garcia-Suarez, J.***, Seylabi, E., & Asimaki, D. “Linear one-dimensional site response analysis in the presence of stiffness-less free surface for certain power-law heterogeneities”, *Soil Dynamics and Earthquake Engineering* (2021), DOI: 10.1016/j.soildyn.2020.106530

• 2. **Garcia-Suarez, J.***, & Asimaki, D. “Exact seismic response of smooth rigid retaining walls resting on stiff soil”, *International Journal for Numerical and Analytical Methods in Geomechanics* (2020), DOI: 10.1002/nag.3082

• 1. **Garcia-Suarez, J.***, & Asimaki, D. “On the fundamental resonant mode of inhomogeneous soil deposits”, *Soil Dynamics and Earthquake Engineering* (2020), DOI: 10.1016/j.soildyn.2020.106190

Peer-reviewed conference articles _____

• 3. **Garcia-Suarez, J.***, & Asimaki, D. “Towards a reliable analytic method for the assessment of seismically-induced pressures on retaining walls”, *Proceedings of the 7th International Conference on Earthquake Geotechnical Engineering* (2019)

• 2. Asimaki, D., **Garcia-Suarez, J.**, Kusanovic, D., Nguyen, K., & Seylabi, E. “Next generation reduced order models for soil-structure interaction”, *Proceedings of the 7th International Conference on Earthquake Geotechnical Engineering* (2019)

• 1. **Garcia-Suarez, J.***, & Asimaki, D. “A quasi-static displacement-based approximation of seismic earth pressures on rigid walls”, *Proceedings of the 5th Geotechnical Earthquake Engineering and Soil Dynamics Conference* (2018)

Research Projects: in Preparation/Submitted

► **European Research Council (ERC)**

Marie Skłodowska-Curie Actions - Staff Exchanges (2026 call). Submitted. Project: “*Multi-scale Mechanics of Advanced Materials under Hypervelocity Impact*” (*HyperMech*). Funding for secondments for four years, to sponsor team member visits (3-month a year max.) to partners (e.g., Princeton University).

Approved Research Projects

► **Swiss National Science Foundation (SNSF)**

Ambizione fellowship (2022 call). Amount: CHF 830,128 (approx. USD 1M). Dates: January 2024 - December 2027. Project: “*Data-driven computational friction*” (*DDCF*). Funding for four-year project, including own’s salary and for hiring one graduate student.

Invited talks

► (12) **Instituto Madrileño de Estudios Avanzados (IMDEA Materials, Madrid)**

Talk titled “Wave propagation regimes in beams”, April 17th, 2026

► (11) **Chair of Structural Mechanics and Monitoring (ETH Zürich)**

Talk titled “Layered media beyond propagators”, November 30th, 2025

► (10) **Department of Mechanical and Aerospace Engineering: MAE Seminar Series (University of Florida)**

Talk titled “Fluid-mediated impact of soft solids”, October 9th, 2025

- ▶ (9) **12th European Solid Mechanics Conference. Mini-symposium “Data driven approaches of materials structures and processes” (Lyon, France)**
Talk titled “Phase-space iterative solvers”, July 10th, 2025
- ▶ (8) **Special Slam Seminar (Institute of Science and Technology Austria)**
Talk titled “Fluid-mediated impact of soft solids”, September 10th, 2024
- ▶ (7) **M2L Workshop (City, University of London)**
Talk titled “Dynamic friction meets machine learning”, June 26th, 2024
- ▶ (6) **TRIPOP seminar (Centre Inria de l’Université Grenoble Alpes)**
Talk titled “Phase-space iterative solvers”, March 15th, 2024
- ▶ (5) **Department of Mechanical Engineering: Guest Series (University of Seville)**
Talk titled “Phase-space iterative solvers”, November 30th, 2023
- ▶ (4) **Instituto Madrileño de Estudios Avanzados (IMDEA Materials, Madrid)**
Talk titled “D-refinement: a new kind of solver for a new paradigm in computational mechanics”, January 9th, 2023
- ▶ (3) **Society of Engineering Science Annual Technical Meeting (College Station, Texas)**
Talk titled “FEM-DEM coupling bridging-zone coupling methods”, October 17th, 2022
- ▶ (2) **Mechanics and Materials Laboratory (ETH Zürich)**
Talk titled “Harmonic structure of 1D transfer matrices”, June 27th, 2022
- ▶ (1) **Knowles Symposium (Caltech)**
Talk titled “Bounding forces on underground structures using the J-integral”, March 6th, 2020

Additionally, other contributed talks at conferences like SWICCOMAS, EMMC, ESMC, SES and EMI (wave propagation, continuum mechanics), ICCCM and CMIS (tribology), ICEGE and GEESD (geotechnical earthquake engineering).

Professional Service

Congress on Numerical Methods in Engineering (CNM 2026) _____

- ▶ Co-organizer of thematic session “Wave Propagation and Associated Inverse Problems”, *Gijon, Spain, July 2026*

International Conference on Advanced Ceramics and Composites (ICACC 2025) _____

- ▶ Co-organizer of symposium “Integrated Computational-Experimental Modeling and Design of Ceramics and Composites”, *Daytona Beach FL, January 2025*

Institutional Service

EPFL’s Doctoral Program in Energy - Ph.D. Candidacy Jury _____

- ▶ Member of candidacy committee of Ph.D. student Mr. Karl Sturm, *December 15th 2025*

EPFL's MEGA (MEchanics GAttering) seminars _____

- ▶ Meeting coordinator (newsletter management), *September 2024 – June 2025*

EPFL's Computational Solid Mechanics Laboratory _____

- ▶ Group meeting coordinator (scheduling talks and hosting external speakers), *February 2022 – September 2023*

Caltech's Graduate Aerospace Laboratories (GALCIT) _____

- ▶ Member of Galcit Graduate Student Council, *Academic year 2018-2019*
- ▶ Member of Hornung Committee for the best thesis delivered by a Galcit student, *Academic year 2017-2018*

Teaching and Mentoring Experience

Main Instructor _____

- ▶ École Polytechnique Fédérale de Lausanne: **CIVIL-425 Continuum mechanics and applications** (spring semester academic year 2024-2025), 6-credit master's level class. Tasks: course development and organization, lecturing. Additional lectures by Profs. J.-F. Molinari and B. Lecampion. Quotes from students:

"I am very thankful for Joaquin to put so much effort into guiding me through this course."

"Prof. Suarez is very dynamic and it is very enjoyable to come to the class."

"Very appreciated that Joaquin remembered our individual name, made us feel closer to [the] teacher and inspired a friendly ambiance."

Teaching Assistant _____

- ▶ École Polytechnique Fédérale de Lausanne: **CIVIL-425 Advanced continuum mechanics** (fall semester academic year 2021-2022), 3-credit master's level class. Tasks: office hours, recitations, lectures (a total of six hours of class, filling in when instructor was not available).
- ▶ California Institute of Technology: **AM/CE 151a Dynamics and vibrations** (Fall 2017, Fall 2018, Fall 2019), 9-unit graduate level class. Tasks: office hours, recitations, grading, lectures (between three and six times each year, whenever instructor not available). Quotes from students:

"Joaquin is a natural teacher, with a deep understanding of the material and a knack for elucidating complexity."

Master and bachelor student projects _____

Summary: 2 master's theses, 7 master's semester projects, 10 bachelor semester projects. Project names marked with * highlight those that led to publications.

- École Polytechnique Fédérale de Lausanne: Mr. **Vibhu Bhaivab** (master's semester project "*DEM study of interaction between a solid body and a granular medium*", fall and spring semester academic year 2021-2022, 12 credits), Mr. **Arthur Cornet** (alongside Sacha Wattel, master's semester project "*Multiscale modeling of 1D site response analysis with data-driven computational mechanics and discrete-element method**", spring semester academic year 2021-2022, 6 credits), Mr. **Louis Jacque** (alongside Manon Voisin-Leprince, master's semester project "*Development of DEM simulations of the interaction between a shoe and soil*", spring semester academic year 2021-2022, 6 credits), Mr. **Marc Toure** (bachelor's semester project "*Outsole-and-soil contact simulation with the FEM*", spring semester academic year 2021-2022, 6 credits), Mr. **Arthur Hattstat** (bachelor's semester project "*Heat transfer in layered media*", spring semester academic year 2021-2022, 6 credits), Mr. **Antoine Binggeli** (alongside Sacha Wattel, bachelor's semester project "*Data-driven constitutive laws using neural networks*", spring semester academic year 2022-2023, 6 credits), Mr. **Paul Croquenois** and Mr. **Alexandre Mascaro** (bachelor's semester project "*Heat transfer in layered media*", spring semester academic year 2022-2023, 6 and 7 credits, respectively), Mr. **Nur C. Sangiorgio** (master's semester project "*Convergence properties of phase-space iterative solvers**", spring semester academic year 2023-2024, 12 credits), Ms. **Yevheniia Tochona** (master's semester project "*Using PINNs to solve friction problems*", fall semester academic year 2024-2025, 12 credits), Mr. **Mathieu Salz** (master's semester project "*Physics-informed neural networks for friction*", spring semester academic year 2024-2025, 12 credits), Mr. **Younis Bouzar** (master's thesis "*Spring-block system: a bifurcation analysis of rate-and-state*", spring semester academic year 2024-2025, 30 credits), Mr. **Nur C. Sangiorgio** (master's thesis "*Wave propagation in phononic crystals under the lens of Lie theory**", fall semester academic year 2025-2026, 30 credits), Mr. **Mathieu Salz** (master's semester project "*From Fast Fourier Transform (FFT) to Fourier Neural Operator (FNO)*", spring semester academic year 2025-2026, 12 credits), Mr. **Ozven Fontani** and Mr. **Jules Champion** (alongside Prof. J.-F. Molinari bachelor's semester project "*Contact mechanics of sport footwear*", spring semester academic year 2025-2025, 6 credits), Mr. **Inacio Vieira** and Ms. **Clémentine Baudou** (alongside Jacopo Bilotto, bachelor's semester project "*Soft lubrication of solid impact*", spring semester academic year 2025-2026, 6 credits).

Research intern supervision _____

- École Polytechnique Fédérale de Lausanne: Mr. **Mathieu Salz** (intern funded by SNSF via Ambizione fellowship, July 2025 – August 2025), research on using neural operators to solve linear and non-linear problems.

Graduate student supervision _____

- École Polytechnique Fédérale de Lausanne: Mr. **Lazar Obradovic** (student funded by SNSF via Ambizione fellowship, March 2024 – June 2024, had to leave the PhD track due to personal reasons). Mr. **Gaëtan Cortes** (student funded by SNSF via Ambizione fellowship, thesis titled "*Data-driven simulations of frictional phenomena*", projected graduation date in 2027, starting date September 2nd 2024).

Graduate student mentoring _____

- ▶ École Polytechnique Fédérale de Lausanne: Dr. **Manon Voisin-Leprince** (alongside Prof. J.-F. Molinari and Dr. G. Anciaux, thesis titled “*Modeling friction and wear using an adaptive discrete-continuum coupling*”, graduated in 2024), Mr. **Sacha Wattel** (alongside Prof. J.-F. Molinari, thesis “*Wear across scales*”, graduated in 2025).

Peer Review _____

Academic journals _____

- ▶ International Journal for Numerical Methods in Engineering (Wiley), Solid State Communications (Elsevier), Mechanics of Materials (Elsevier), Tribology Letters (Springer Nature), Journal of Applied Physics (American Institute of Physics), Soil Dynamics and Earthquake Engineering (Elsevier), International Journal for Numerical and Analytical Methods in Geomechanics (Wiley), Journal of Geotechnical and Geoenvironmental Engineering (American Society of Civil Engineers)

Scientific Societies Membership _____

- ▶ Society of Engineering Science (SES)
- ▶ ASCE’s Engineering Mechanics Institute (EMI)
- ▶ International Union of Theoretical and Applied Mechanics (IUTAM)
- ▶ European Mechanics Society (EuroMech)
- ▶ Spanish Society for Theoretical and Applied Mechanics (SEMTA)

Awards _____

Prizes _____

- ▶ **Demetriades-Tsafka-Kokkalis Prize in Seismo-Engineering, Prediction, and Protection**, Academic year 2019-2020
- ▶ **Registrar’s Office Excellent TA Mention** (based on results of student survey), Fall 2017

Fellowships _____

- ▶ **Ambizione Fellowship** – Swiss National Science Foundation, 2023
- ▶ **Tyson Graduate Fellowship** – Mechanical & Civil Engineering Department (Caltech), 2017
- ▶ **Talentia Fellowship** – Andalusia Regional Government, 2015

Skills _____

- ▶ **Core technical competences:** applied mathematics (modeling, PDEs, numerical analysis, group theory), continuum mechanics (both solid and fluid mechanics)
- ▶ **Programming:** C++, Python, Wolfram Mathematica
- ▶ **Languages:** English (proficient), French (intermediate), Spanish (native)

Social Media

🐦 @jgs_research: outreach platform for sharing research updates.